

ELECTRICAL SYSTEMS & CONTROLS ENGINEERING PORTFOLIO

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(All systems were designed, tested, and documented in accordance with Caltrans standards, including applicable TEES requirements, as well as the National Electrical Code (NEC) and the California Electrical Code.)

Welcome and thank you for taking the time to view my portfolio. The Goal of this portfolio is to give you a deeper insight into my experiences and skills I have gained over my career history.

It is my hope that this will allow you to better assess how my skills can be applied to your company.

BILL OF MATERIALS (BOM) OWNERSHIP, VENDOR OPTIMIZATION & PROCUREMENT LIFECYCLE MANAGEMENT

Role:

Electrical / Systems Engineer & Senior Production Lead
(Acting Technical Business Partner for Procurement and Operations)

Overview:

Served as the primary technical owner for Bills of Materials (BOMs) across complex infrastructure deployments, bridging engineering intent with procurement execution and vendor manufacturing realities. Frequently assigned incomplete or problematic BOMs due to demonstrated ability to resolve ambiguities, identify hidden dependencies, and convert under-defined documentation into procurement-ready, execution-safe deliverables that reduced risk, delays, and cost overruns.

BOM Governance & Ownership:

Took ownership of BOMs that were initially vague, under-scoped, or missing critical components, quantities, and accessories required for production and field deployment. Expanded these BOMs into complete, execution-ready documents aligned with real-world installation, commissioning, and maintenance needs. Maintained BOMs as living documents, incorporating late-stage design changes, field-driven additions, and procurement feedback while preserving version control, traceability, and auditability across project lifecycles.

Vendor Optimization & Part Strategy:

Managed detailed vendor mappings within BOMs, including manufacturer references, operational part numbers, procurement identifiers, compatibility notes, and approved alternates. Worked directly with equipment suppliers to validate part availability, confirm substitutions, and obtain updated pricing and lead-time information, ensuring procurement and project teams had accurate, current data to support purchasing decisions.

Proactively re-engaged suppliers to optimize legacy procurement specifications that introduced unnecessary cost, extended lead times, or added production burden. Identified cases where delivered assemblies included excess features routinely removed during operational preparation and collaborated with supplier engineering teams to redesign those assemblies for improved manufacturability and operational efficiency.

Led a redesign effort for standardized roadside enclosures that were consistently overbuilt for the intended application. Simplified operational layouts, removed nonessential components, and updated procurement specifications to better reflect actual operational needs. These changes reduced preparation time for production teams and lowered per-unit production costs.

Initiated a material and finishing process change by replacing anodized finishes with powder coating, eliminating reliance on third-party finishing vendors. This reduced environmental exposure, shortened manufacturing lead times, and improved delivery predictability. The redesign lowered cost and complexity for both project operations and suppliers and was incorporated into updated BOMs to ensure long-term adoption without recurring rework.

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BILL OF MATERIALS (BOM) OWNERSHIP, VENDOR OPTIMIZATION & PROCUREMENT LIFECYCLE MANAGEMENT

End-of-Life (EOL) Risk Mitigation:

Designed and implemented an operational EOL awareness procedure to reduce procurement disruptions caused by discontinued components. Proactively contacted key vendors to understand how EOL notifications were issued, typical advance notice periods, and available substitution pathways. Translated this information into actionable guidance for procurement teams, enabling early detection of EOL risks and preventing last-minute redesigns, emergency purchases, or field delays.

Business Impact:

- Reduced procurement delays caused by incomplete or inaccurate BOMs
- Minimized rework and field installation issues due to missing or unnecessary components
- Lowered production preparation time through vendor-driven design simplification
- Improved cost predictability through early vendor engagement and quote validation
- Shortened lead times by eliminating third-party finishing dependencies
- Increased organizational resilience to EOL events and supply chain disruptions
- Established trust as a go-to technical partner for high-risk or high-visibility BOMs

Positioning Statement:

Functioned as a technical business partner between engineering, procurement, vendors, and operations, ensuring that design intent translated into executable, cost-aware, and lifecycle-resilient procurement strategies while actively improving vendor designs to reduce cost, risk, and production friction.

TEMPORARY ELECTRONIC TOLLING SYSTEM

Client:

Regional Transportation Authority (Northern California)

Role / Company:

Electrical & Systems Engineer – Transportation Systems Integrator

Project Overview:

Contributed to early-stage system design and integration for an interim electronic tolling system deployed under strict spatial constraints. The project required fitting all power, communication, and control subsystems into a limited number of existing roadside cabinets with no physical expansion options, while maintaining serviceability and regulatory compliance.

Technical Approach:

Developed space-efficient cabinet layouts through detailed power load analysis and consolidation of redundant power supplies. Produced precise electrical layouts and documentation using industry-standard CAD and drafting tools, ensuring compliance with applicable electrical codes and transportation standards. Collaborated closely with production, field, and network teams to validate cabinet configurations and overall system compatibility.

Key Contributions:

Performed voltage drop and distance calculations to optimize wire routing and conduit sizing. Updated and validated the bill of materials to reflect current components, availability, and cost considerations. Delivered a balanced cabinet design that integrated power redundancy, safety requirements, and accessibility within fixed enclosure constraints.

Results and Impact:

Enabled a compliant, space-optimized system architecture that preserved full power redundancy while simplifying maintenance and installation workflows. Improved coordination between engineering, production, and installation teams, and established a reusable design framework for future tolling and roadside infrastructure projects.

Key Learnings:

Strengthened expertise in early-stage system planning under physical and regulatory constraints. Expanded proficiency with electrical design tools and industry codes. Reinforced the importance of cross-functional collaboration to resolve complex infrastructure challenges efficiently.

CONTROL CABINET DESIGN, BUILDOUT, AND COMMISSIONING

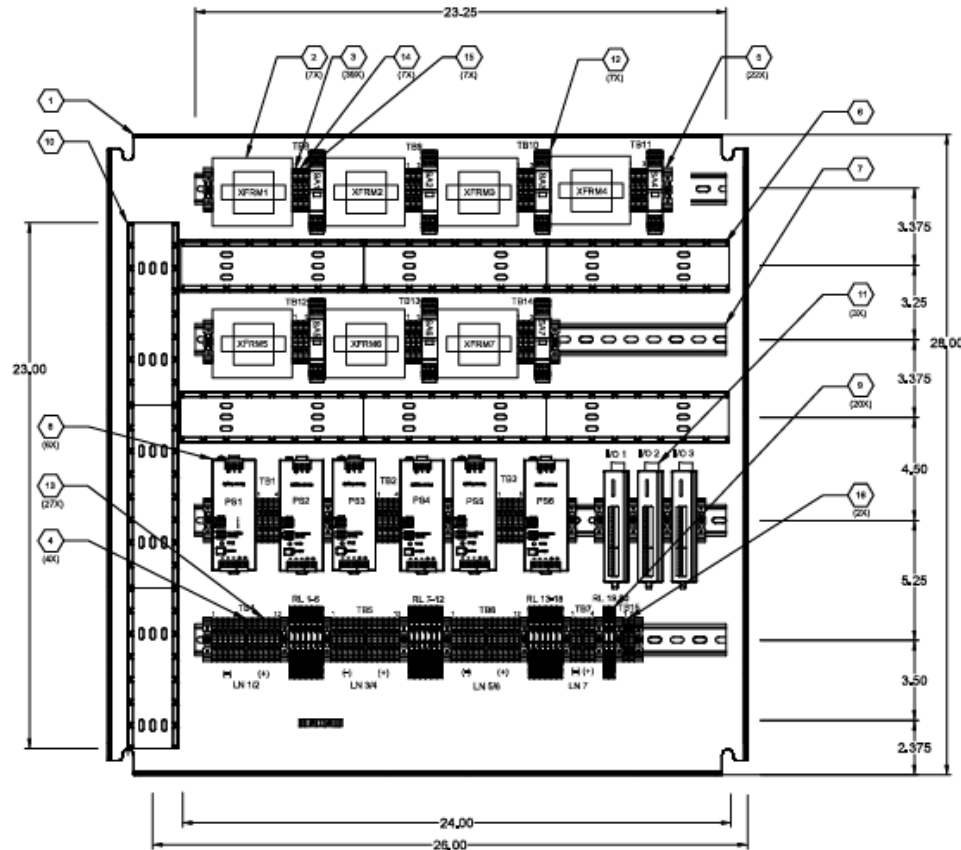
This section presents representative control cabinet designs I owned from system definition through buildout, testing, and field commissioning. The examples shown reflect real, deployed infrastructure systems and illustrate how design intent was translated into executable, serviceable hardware under regulatory, spatial, and operational constraints.

My responsibilities spanned electrical architecture, component selection, power distribution strategy, PLC and I/O integration, wiring and grounding design, and documentation sufficient for production and field execution without engineering rework. Each cabinet was designed to meet NEC and CEC requirements, support reliable operation in harsh environments, and enable deterministic behavior during normal operation, power transitions, and fault conditions.

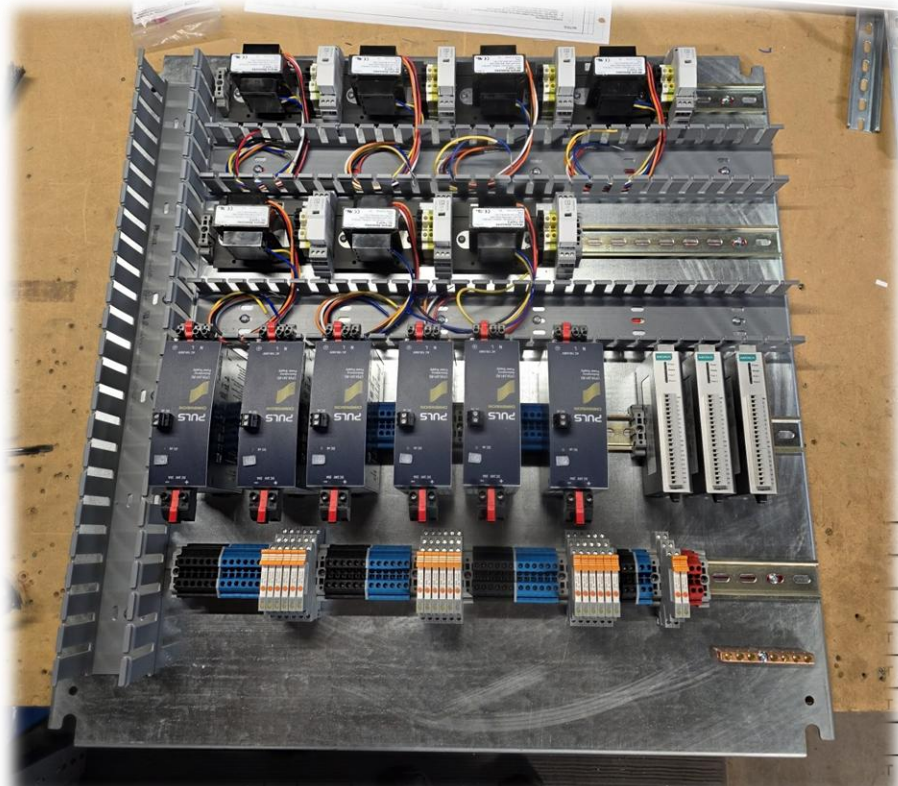
The drawings and photos that follow are included to demonstrate engineering judgment, layout decisions, and design-for-installation considerations rather than to exhaustively document every system. They highlight how constraints such as enclosure limits, power quality, redundancy, serviceability, and lifecycle maintenance were addressed in practical, buildable designs.

NOTES:

- 1. THIS DRAWING SERVES AS A GENERAL ARRANGEMENT DRAWING AND DOES NOT INCLUDE MISC. HARDWARE FOR ASSEMBLY. EQUIPMENT LOCATION DIMENSIONS ARE APPROXIMATELY AND SHALL BE ADJUSTED AS NECESSARY.
- 2. TOLL SYSTEMINTERGATOR'S ELECTRICAL EQUIPMENT SHALL COMPLY WITH CALTRANS STANDARD SPECIFICATION 88-1.02, "REGULATIONS & CODE".
- 3. ALL DIMENSIONS ARE IN INCHES.
- 4. SEE ED-41 THROUGH ED-43 FOR WIRING DIAGRAM



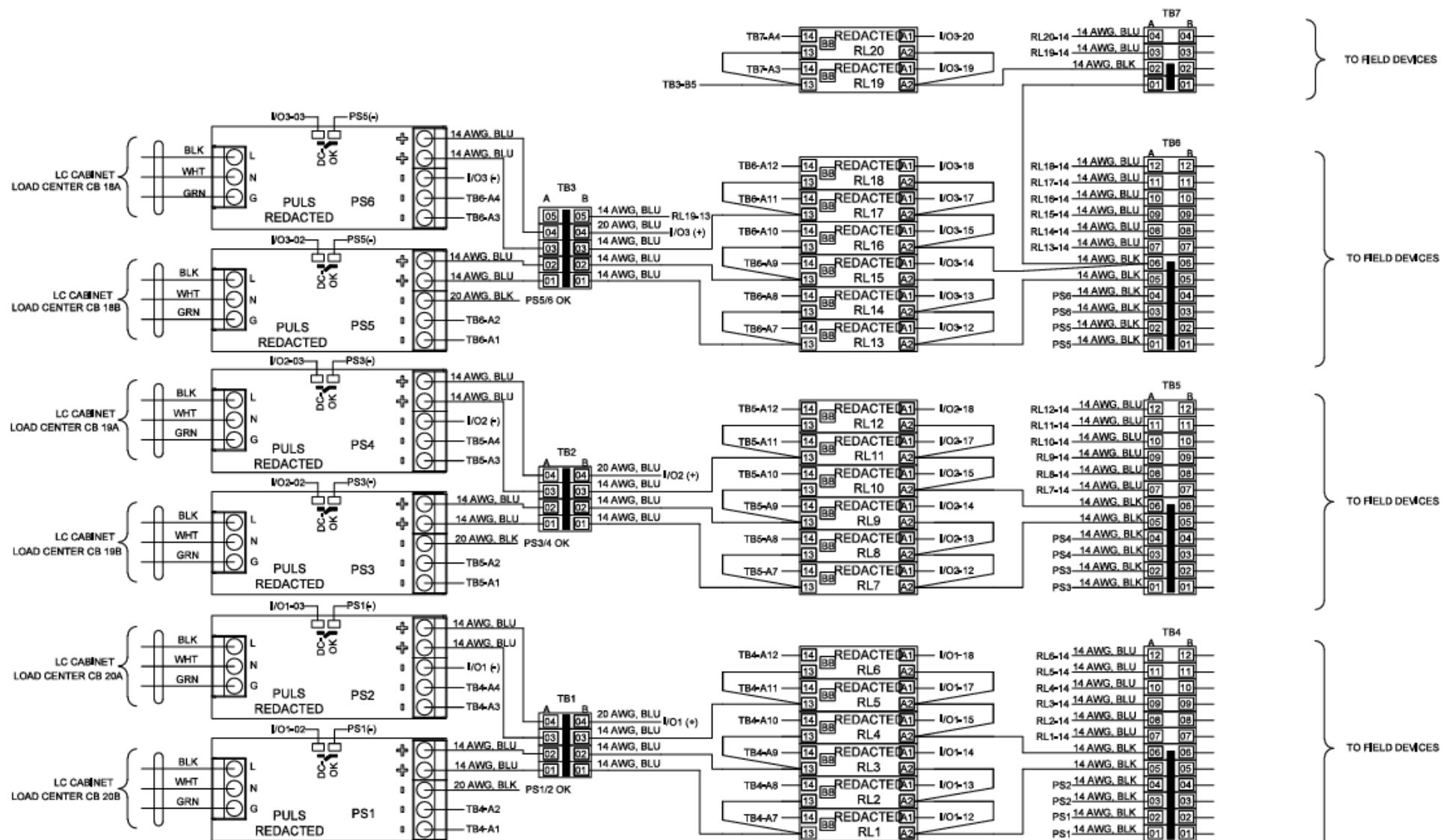
ASSEMBLY DETAIL, POLEMOUNT CABINET



					ITEM
1					1
2					2
3					3
4					4
5					5
6					6
7					7
8			24VDC, 20A POWER SUPPLY	PULS	8
20			PLC-RSC- 24DC/1MS/ACT	PHOENIX CONTACT	9
A/R			2.5" 3" WIRE DUCT	PANDUIT	10
3			UNIVERSAL I/O, 8 DIS, 8 DIOS	MOXA	11
7			24V SURGE ARRESTER	PHOENIX CONTACT	12
33			UT4 TERMINAL BLOCK BLU	PHOENIX CONTACT	13
7			UT4 TERMINAL BLOCK WHT	PHOENIX CONTACT	14
7			UT4 TERMINAL BLOCK PE	PHOENIX CONTACT	15
2			UT4 TERMINAL BLOCK RED	PHOENIX CONTACT	16

NOTES:

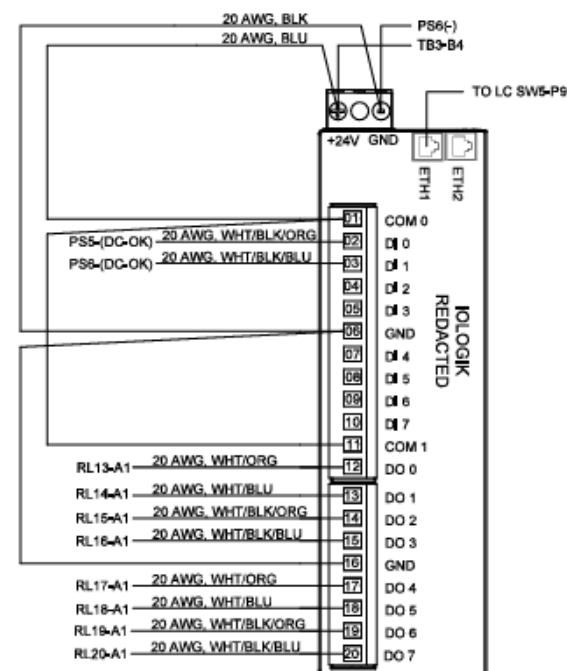
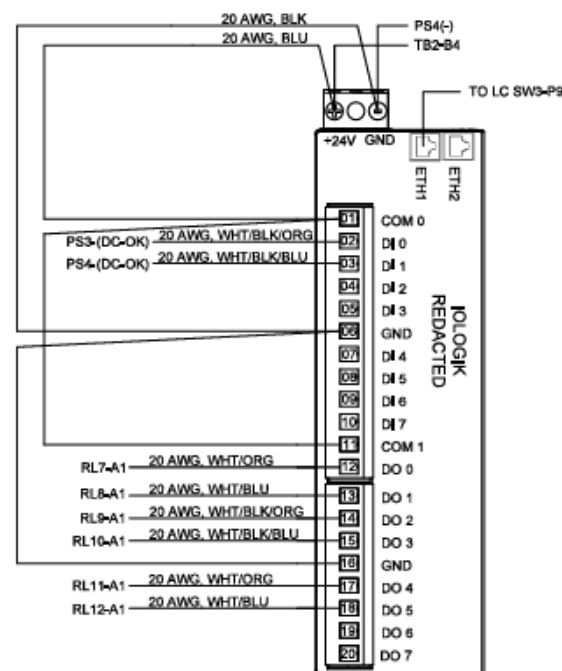
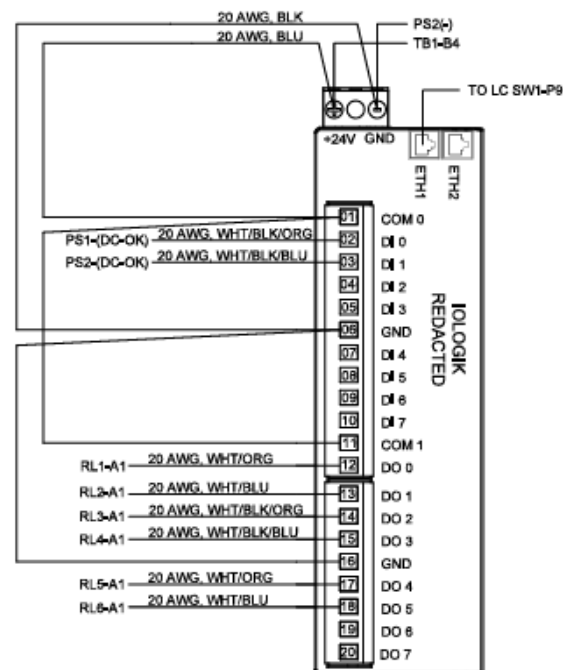
1. ELECTRICAL SYSTEM DETAILS FOR GP CABINET CONTINUE ON ED-42
2. FOR FIELD DEVICE WIRING DIAGRAM SEE ED-58 THROUGH ED-70



ELECTRICAL SYSTEM DETAILS
(WIRING DIAGRAM, GP CABINET)

NOTES:

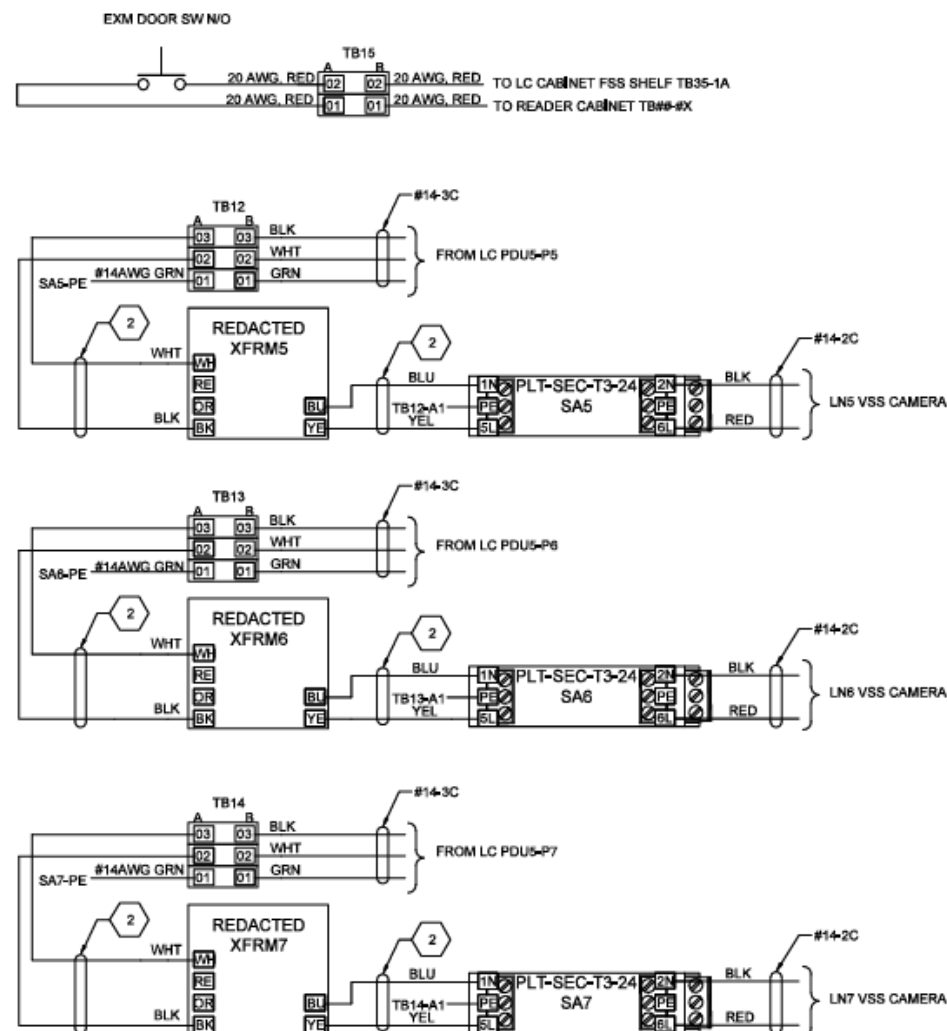
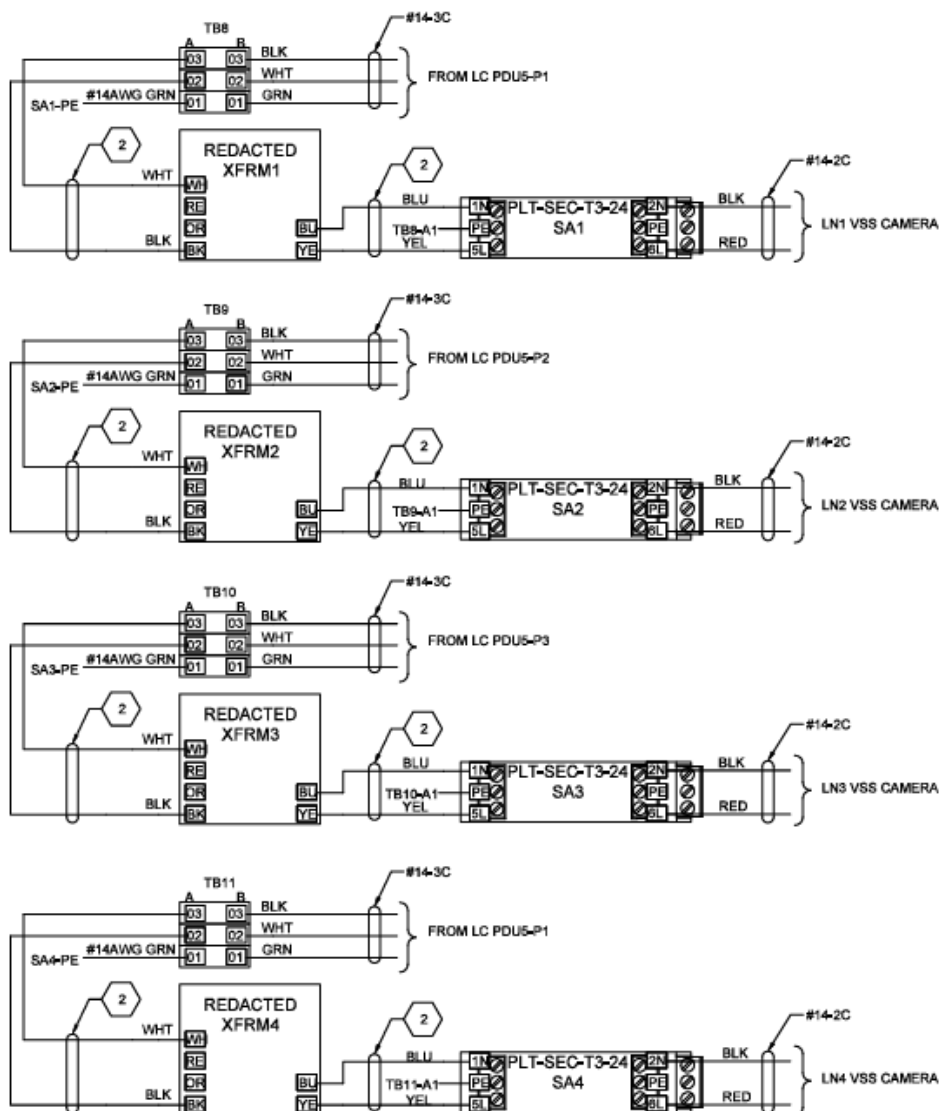
1. ELECTRICAL SYSTEM DETAILS FOR GP CABINET CONTINUE ON ED-43
2. FOR FIELD DEVICE WIRING DIAGRAM SEE ED-58 THROUGH ED-70



NOTES:

1. FOR FIELD DEVICE WIRING DIAGRAM SEE ED-58 THROUGH ED-70

2. CONDUCTORS PROVIDED WITH TRANSFORMER

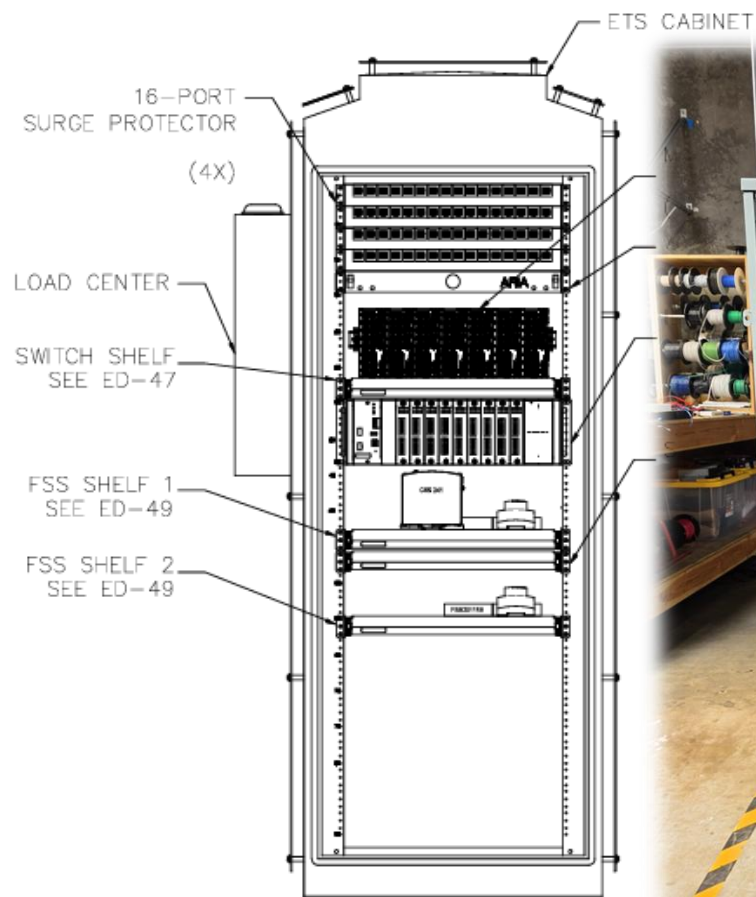


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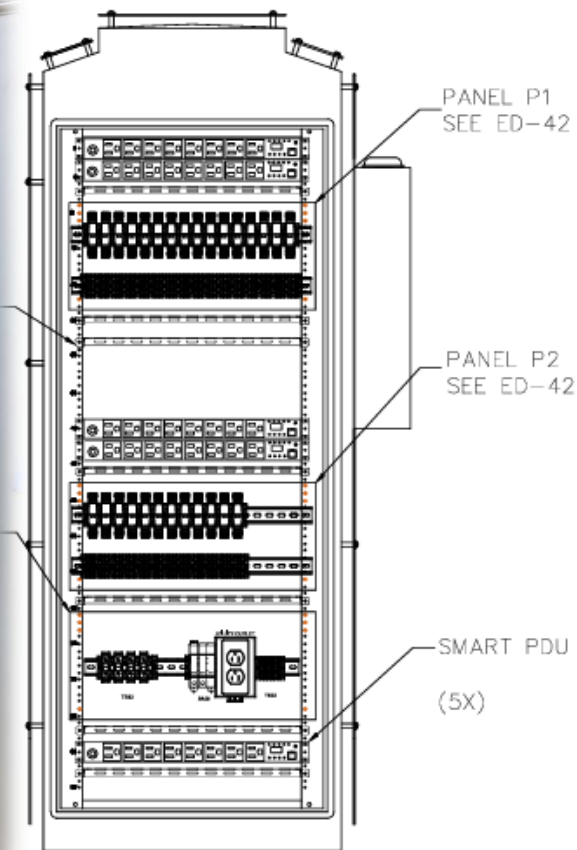
1. THIS DRAWING SERVES AS A GENERAL ARRANGEMENT DRAWING AND DOES NOT INCLUDE MISC. HARDWARE FOR ASSEMBLY. EQUIPMENT LOCATION DIMENSIONS ARE APPROXIMATELY AND SHALL BE ADJUSTED AS NECESSARY.
2. TOLL SYSTEM INTEGRATOR'S ELECTRICAL EQUIPMENT SHALL COMPLY WITH CALTRANS STANDARD SPECIFICATION 89-1.02, "REGULATIONS & CODE".
3. ALL DIMENSIONS ARE IN INCHES.
4. SEE ED-54 THRU ED-55 FOR DETAILS ON LOAD CENTER AFFIXATION

5. TO WIRE AND INSTALL THE IDRIS COMPUTER, ACCESS THE BACK WHILE MOUNTED OR:
 1. PULL THE IDRIS COMPUTER OUT OF THE RACK.
 2. SET IDRIS ON THE EMPTY SLIDING UTILITY SHELF.
 3. ENSURE THERE IS AMPLE SLACK IN THE LOOP HOME-RUNS BEFORE MAKING CONNECTIONS
 4. MAKE CONNECTIONS AND REPLACE IDRIS IN THE RACK

IMPORTANT INSTALLATION NOTE FOR IDRIS



FRONT VIEW



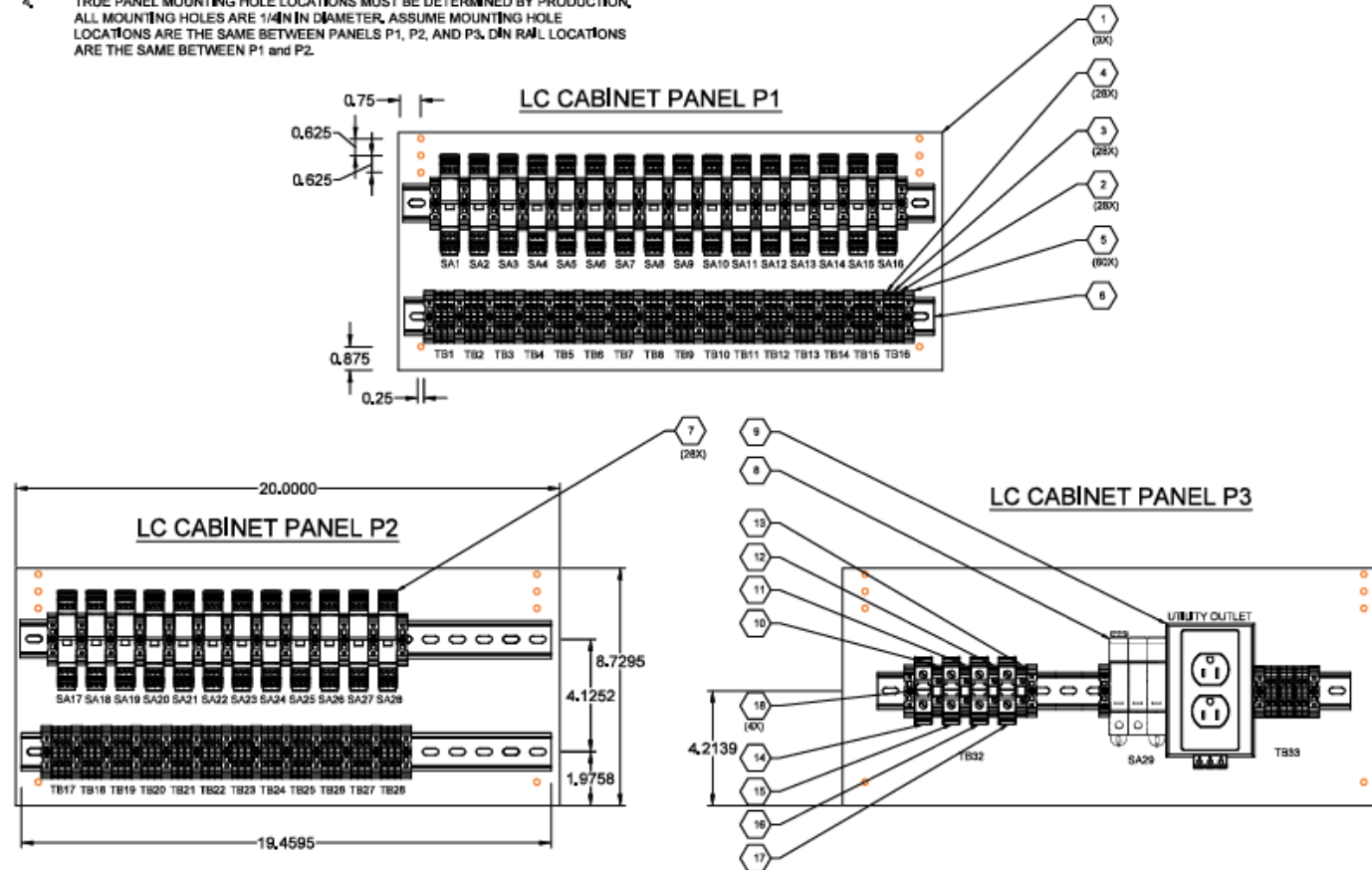
REAR VIEW

ELECTRICAL SYSTEM DETAILS (ASSEMBLY, LC CABINET)

NO SCALE

NOTES:

1. THIS DRAWING Serves AS A GENERAL ARRANGEMENT DRAWING AND DOES NOT INCLUDE MISC. HARDWARE FOR ASSEMBLY. EQUIPMENT LOCATION DIMENSIONS ARE APPROXIMATELY AND SHALL BE ADJUSTED AS NECESSARY.
2. TOLL SYSTEM INTERGATOR'S ELECTRICAL EQUIPMENT SHALL COMPLY WITH CALTRANS STANDARD SPECIFICATION 86-1.02, "REGULATIONS & CODE".
3. ALL DIMENSIONS ARE IN INCHES.
4. TRUE PANEL MOUNTING HOLE LOCATIONS MUST BE DETERMINED BY PRODUCTION. ALL MOUNTING HOLES ARE 1/4 IN IN DIAMETER. ASSUME MOUNTING HOLE LOCATIONS ARE THE SAME BETWEEN PANELS P1, P2, AND P3. DIN RAIL LOCATIONS ARE THE SAME BETWEEN P1 and P2.
5. FOR LC PANEL P1 WIRING DIAGRAM SEE ED-44
6. FOR LC PANEL P2 WIRING DIAGRAM SEE ED-45
7. FOR LC PANEL P3 WIRING DIAGRAM SEE ED-46



ASSEMBLY DETAIL, LC CABINET PANELS

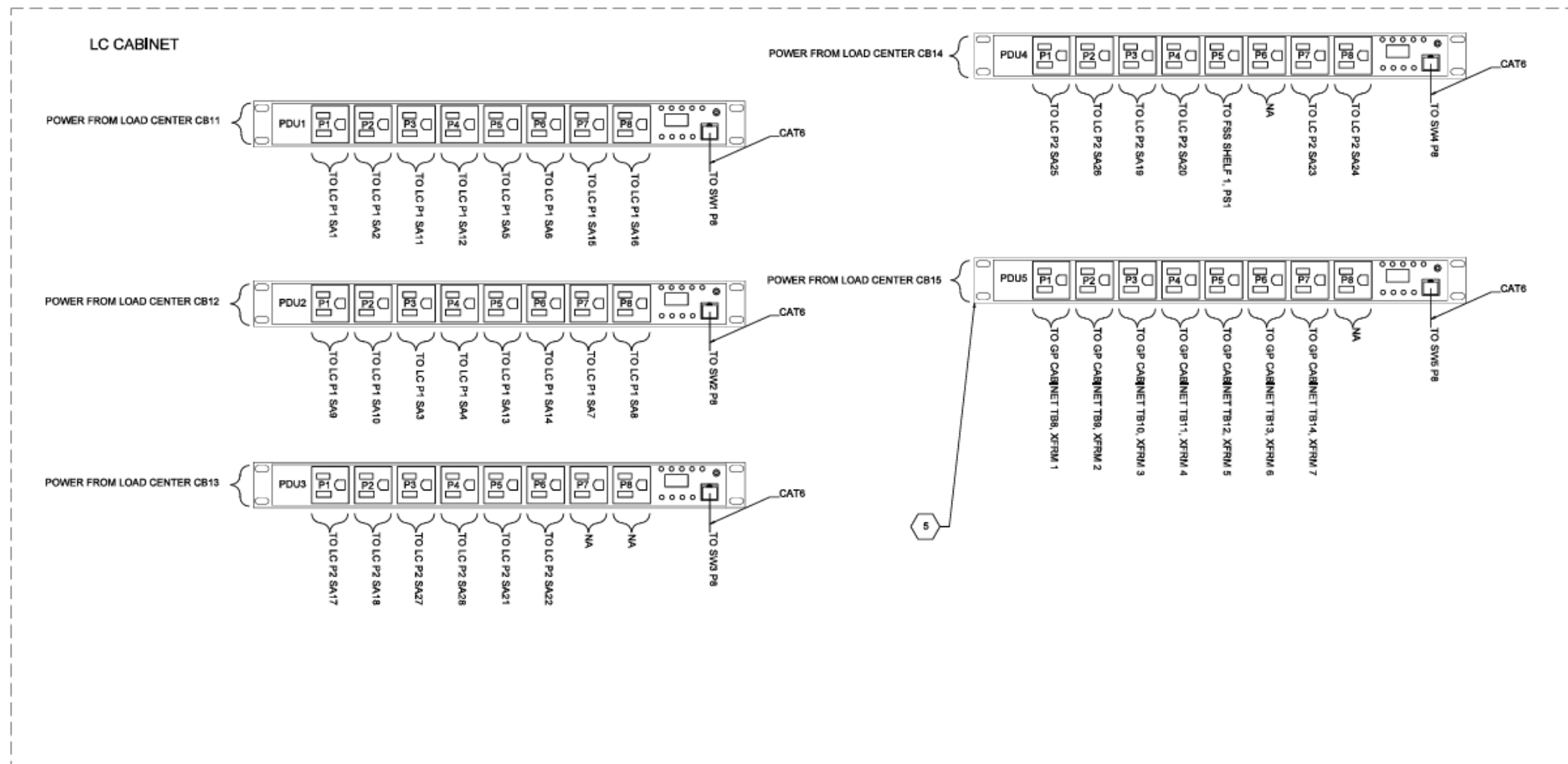
QTY	CODE	PART NUMBER	DESCRIPTION	MANUFACTURE	ITEM
3			20"X8.5"X1/8" PANEL 1, 2 and 3	ALL STATES	1
28			UT4 TERMINAL BLOCK WHT	PHOENIX CONTACT	2
28			UT4 TERMINAL BLOCK PE	PHOENIX CONTACT	3
28			UT4 TERMINAL BLOCK BLK	PHOENIX CONTACT	4
62			END BRACKET	PHOENIX CONTACT	5
A/R			DIN RAIL	PHOENIX CONTACT	6
28			120V SURGE ARRESTER	PHOENIX CONTACT	7
1			POWER IN SURGE PROTECTOR	LITTLEFUSE INC.	8
1			EM-DUO 120/15 - SOCKET	PHOENIX	9
1			UT10 TERMINAL BLOCK BLK	PHOENIX CONTACT	10
1			UT10 TERMINAL BLOCK RED	PHOENIX CONTACT	11
1			UT10 TERMINAL BLOCK WHT	PHOENIX CONTACT	12
1			UT10 TERMINAL BLOCK PE	PHOENIX CONTACT	13
1			UT35 TERMINAL BLOCK BLK	PHOENIX CONTACT	14
1			UT35 TERMINAL BLOCK RED	PHOENIX CONTACT	15
1			UT35 TERMINAL BLOCK WHT	PHOENIX CONTACT	16
1			UT35 TERMINAL BLOCK PE	PHOENIX CONTACT	17
4			RB UT 35-10 - REDUCING BRIDGE	PHOENIX CONTACT	18

NOTES:

1. ALL PDU'S RECEIVE POWER FROM THE LOAD CENTER, SEE ED-52
2. PDU'S NUMBERED FROM TOP OF CABINET TO BOTTOM, 1 THROUGH 7, SEE ED-41
3. PDU POWER CABLES SUPPLIED WITH PDU
4. UNLESS OTHERWISE SPECIFIED, ALL CABLES LEAVING THE PDU'S ARE #14AWG, 3 CONDUCTOR POWER CORDS. USE P/N: INDUSTRIES INC 3-WIRE APPLIANCE AND POWER TOOL CORD, 3 FT, 14 AWG, 15A/125V AC, 1875W

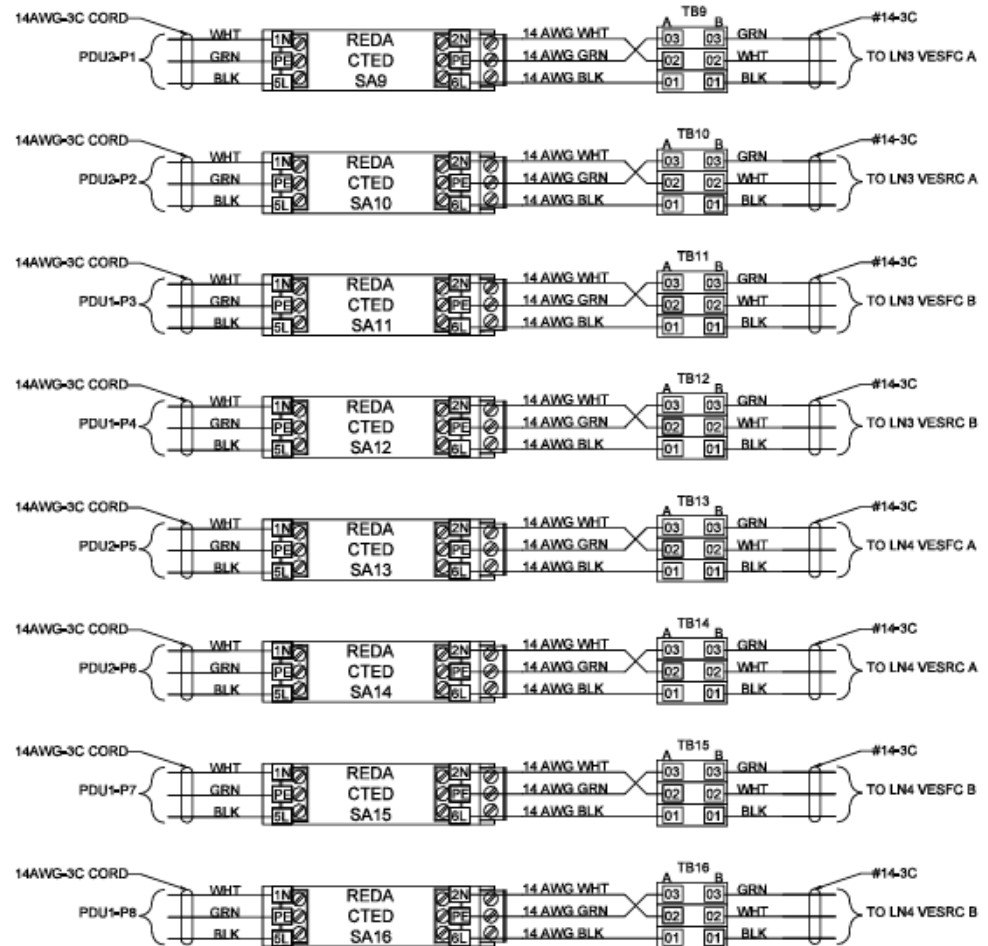
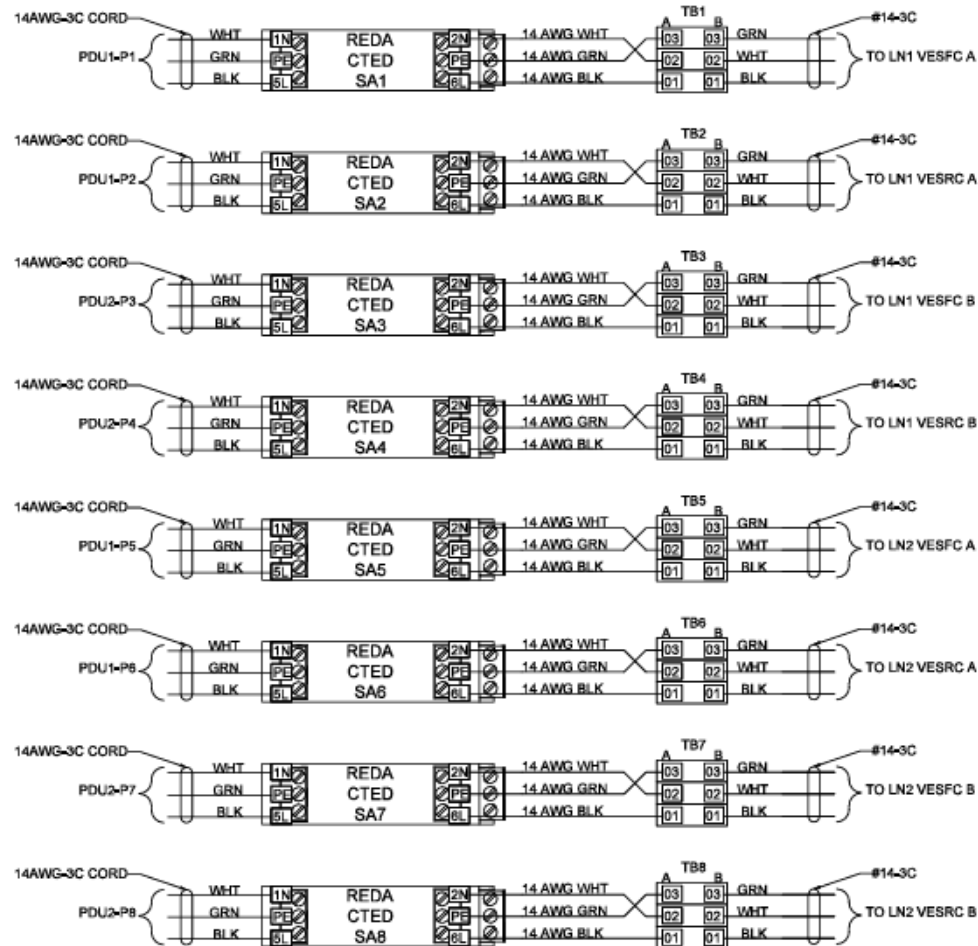
5

UNLESS OTHERWISE SPECIFIED, ALL CABLES LEAVING THESE PDU'S ARE #14AWG, 3 CONDUCTOR CABLE WITH PLUG TO BE FURNISHED BY INSTALLATION TEAM



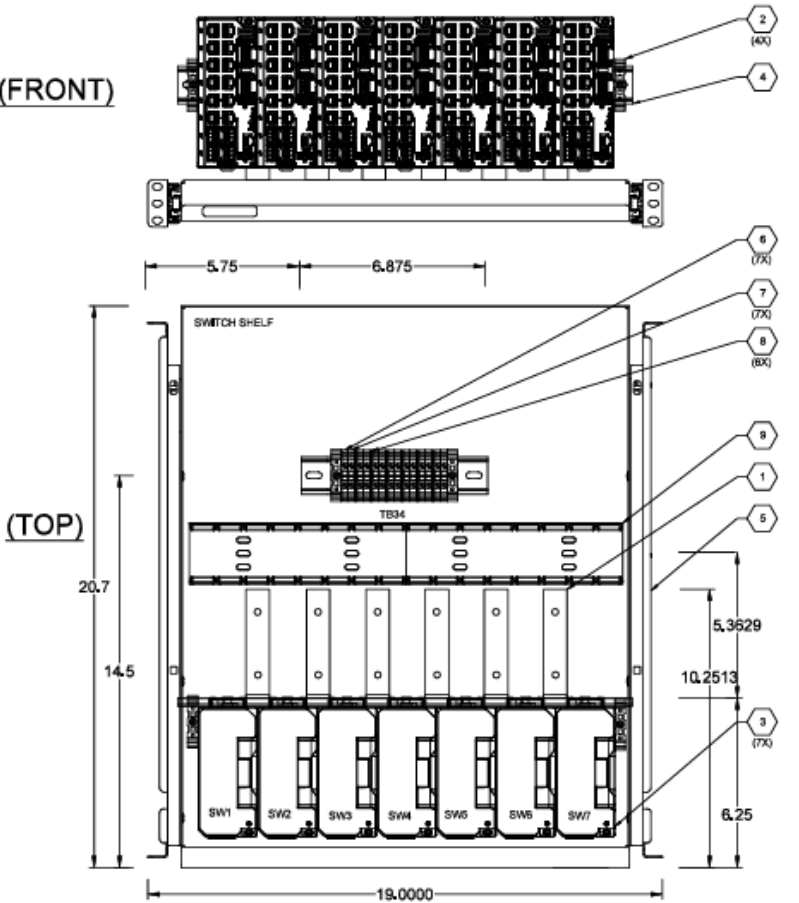
NOTES:

1. FOR 14AWG, 3C CORD
2. WIRING DIAGRAM CONTINUES ON ED-48
3. FOR PANEL P1 LAYOUT, SEE ED-45

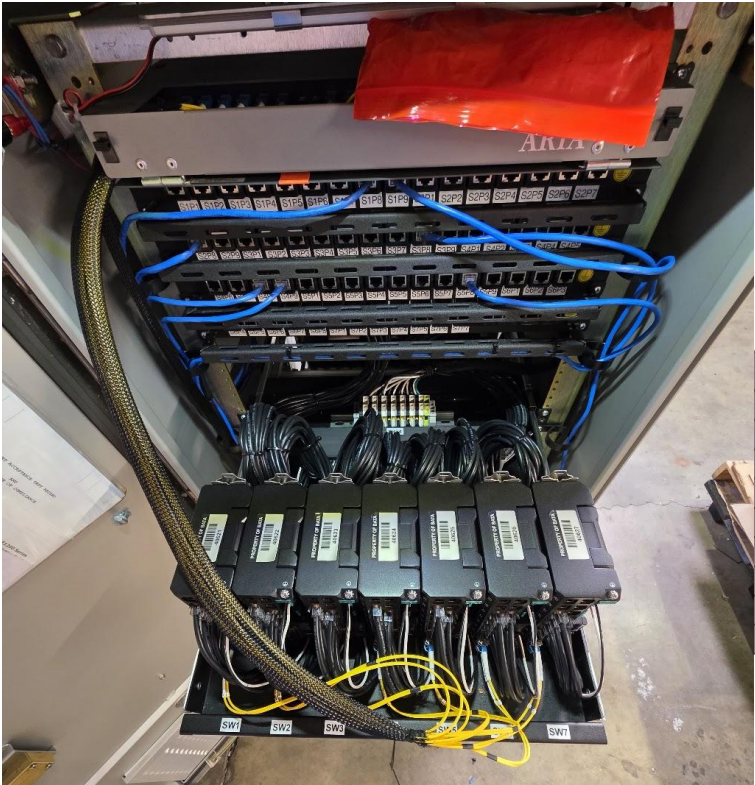


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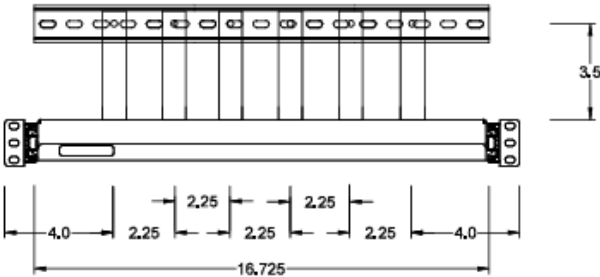
- 1. THIS DRAWING SERVES AS A GENERAL ARRANGEMENT DRAWING AND DOES NOT INCLUDE MISC. HARDWARE FOR ASSEMBLY. EQUIPMENT LOCATION DIMENSIONS ARE APPROXIMATELY AND SHALL BE ADJUSTED AS NECESSARY.
- 2. TOLL SYSTEMINTERGATOR'S ELECTRICAL EQUIPMENT SHALL COMPLY WITH CALTRANS STANDARD SPECIFICATION 66-1.02, "REGULATIONS & CODE".
- 3. ALL DIMENSIONS ARE IN INCHES.
- 4. SEE ED-51 FOR WIRING DIAGRAM



ASSEMBLY DETAIL, LC CABINET SWITCH SHELF

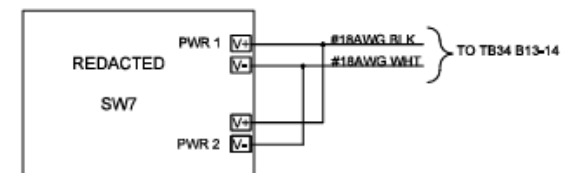
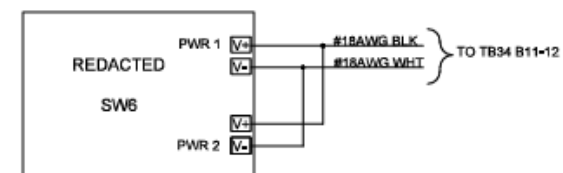
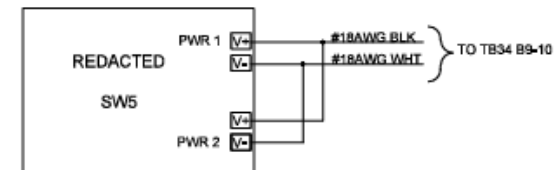
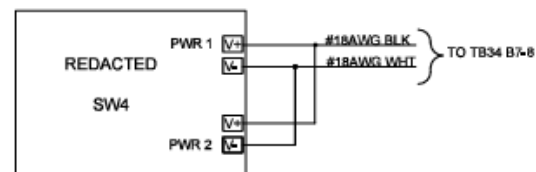
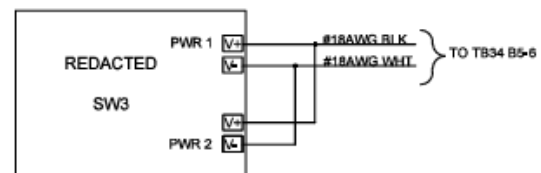
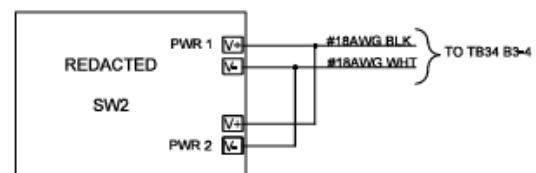
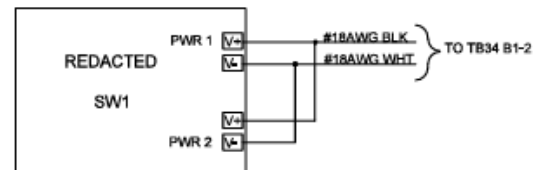
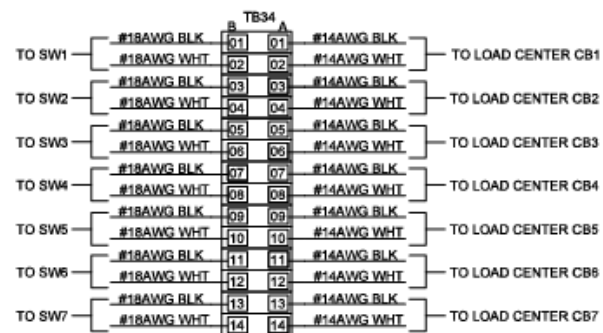


(FRONT: L-BRACKET PLACEMENT)



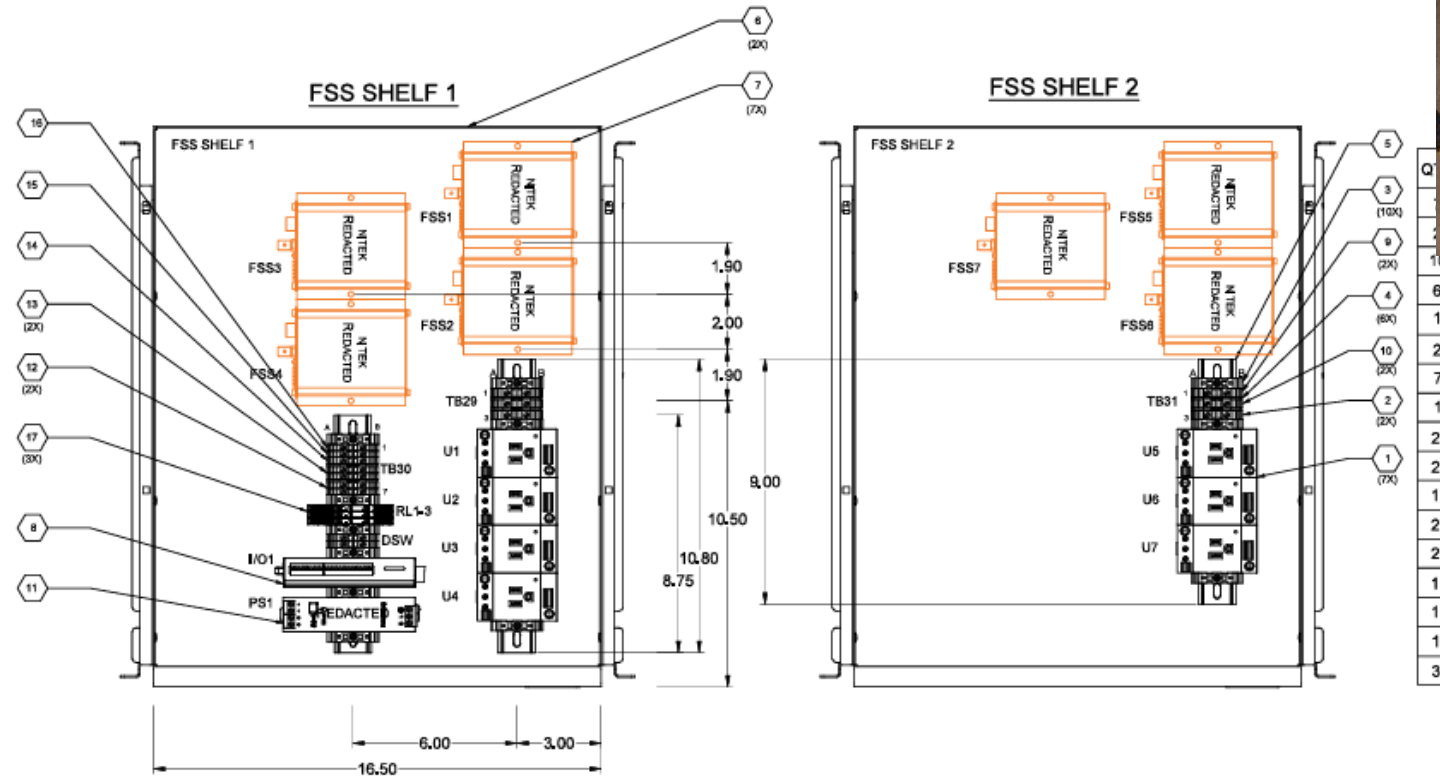
NOTES:

1. FOR SWITCH SHELF LAYOUT SEE ED-50

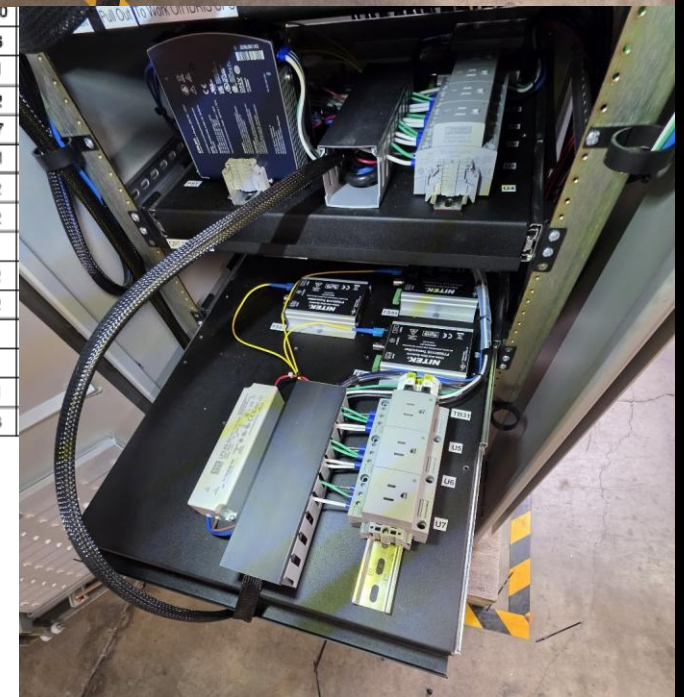
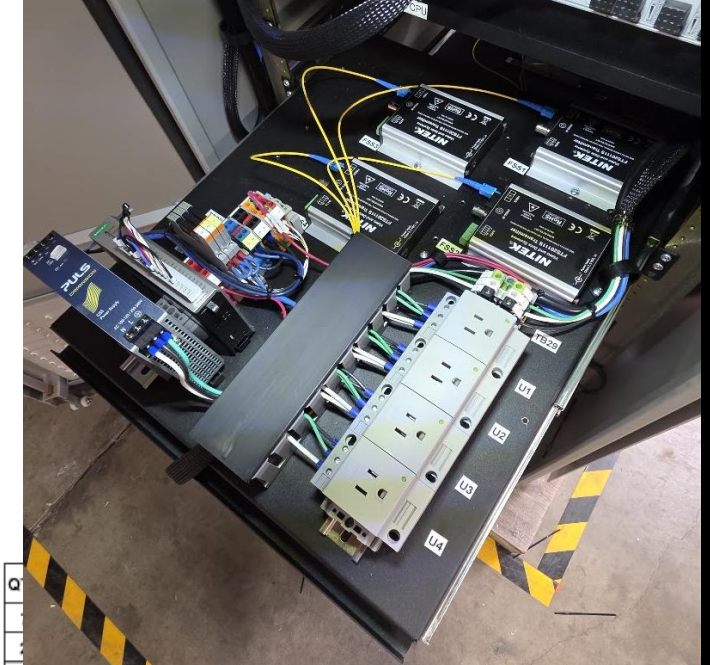


NOTES:

1. THIS DRAWING SERVES AS A GENERAL ARRANGEMENT DRAWING AND DOES NOT INCLUDE MSC. HARDWARE FOR ASSEMBLY. EQUIPMENT LOCATION DIMENSIONS ARE APPROXIMATE AND SHALL BE ADJUSTED AS NECESSARY.
2. TOLL SYSTEM/INTERGATOR'S ELECTRICAL EQUIPMENT SHALL COMPLY WITH CALTRANS STANDARD SPECIFICATION 86-1.02, "REGULATIONS & CODE".
3. ALL DIMENSIONS ARE IN INCHES.
4. SEE ED-53 FOR WIRING DIAGRAM



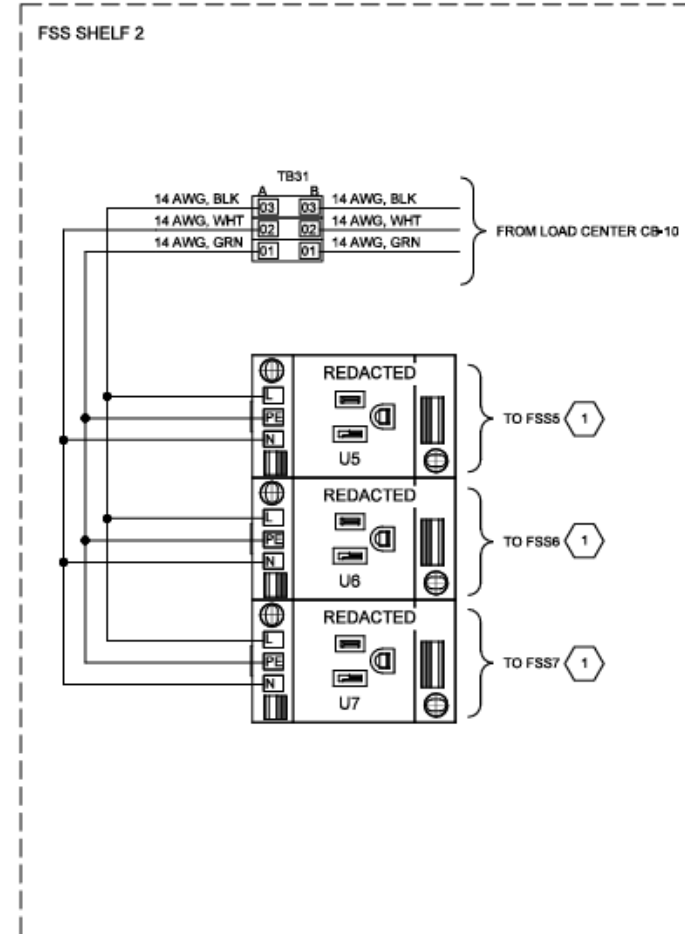
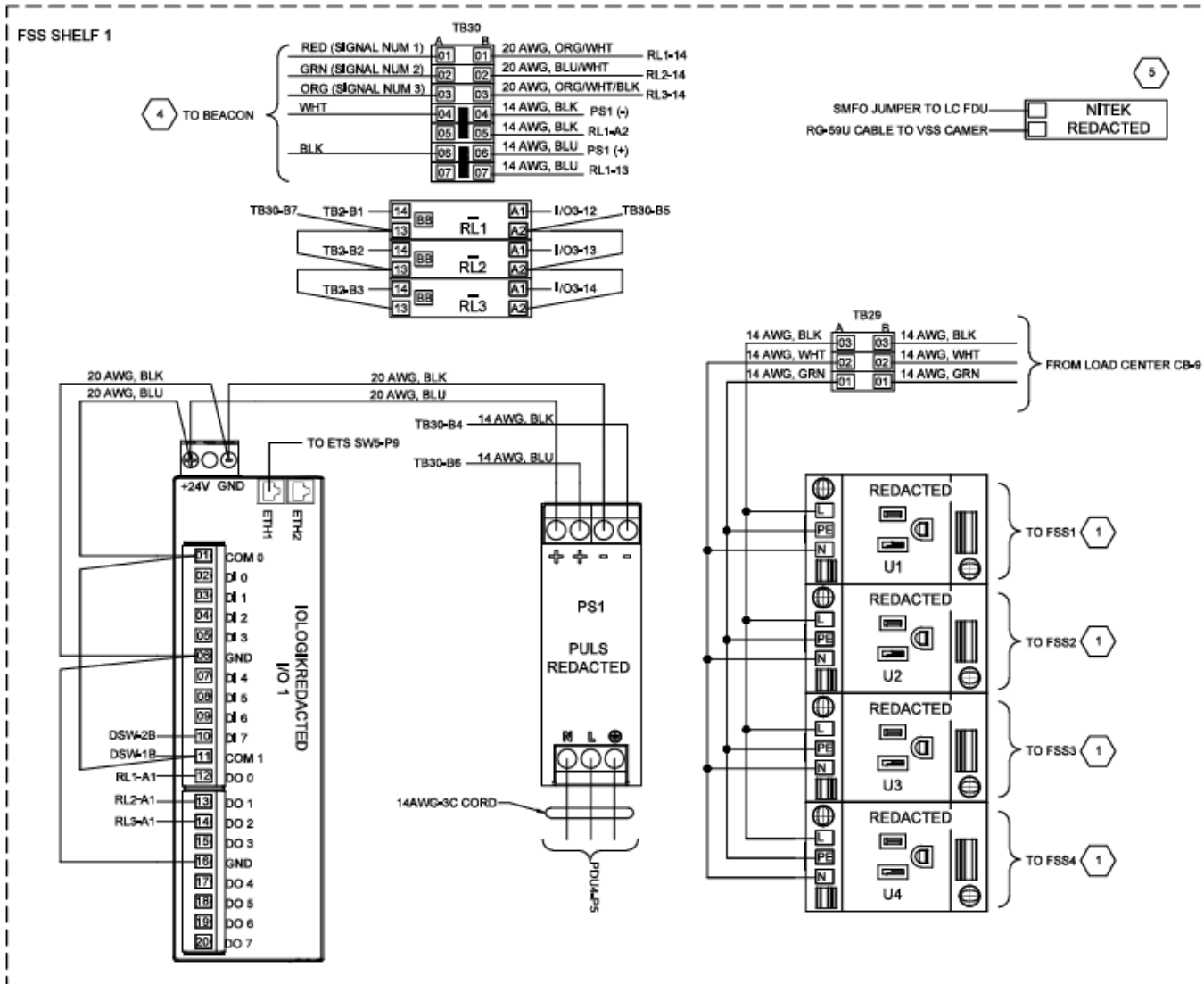
ASSEMBLY DETAIL, LC CABINET PANELS



(ASSEMBLY, LC CABINET FSS SHELVES)

NOTES:

1. USE PROVIDED POWER SUPPLY FROM NITEK REDACTED(MAY CHANGE TO OTHER 5VDC PS)
2. FOR 14AWG, 3C CORD
3. FOR LC FSS SHELF LAYOUT, SEE ED-52
4. BEACON CABLE
5. ALL NITEK FSS MODULE TO BE WIRED THIS WAY.



REMOTE SOLAR-POWERED DETECTION SYSTEM

Client:

Regional Transportation Authority (California)

Role:

Electrical & Systems Engineer – Transportation Systems Integrator

Project Overview:

Designed an autonomous, off-grid solar power system to support a remote vehicle detection and wireless communications installation. The system was required to operate continuously for an extended standalone period while supplying power to sensing and radio transmission equipment in a roadside environment.

Technical Approach:

Developed a custom bill of materials and performed detailed power and energy modeling to determine average load demand, peak draw, and required energy storage. Applied solar resource modeling using location-adjusted peak sun hours and analytical methods derived from formal engineering coursework. Used spreadsheet-based simulations to verify system sizing and operational margins.

System Design:

Specified a photovoltaic-based power architecture incorporating energy storage and charge regulation to meet runtime and reliability constraints. Validated system configuration and sizing through technical coordination with external suppliers, ensuring alignment with availability, performance expectations, and deployment requirements.

Results and Impact:

Delivered a cost-effective, field-ready solar power solution that met multi-day autonomy requirements under full operational load. Achieved first-pass design validation with no required revisions, completing analysis and system definition on an accelerated timeline.

Key Learnings:

Strengthened independent system design and verification workflows. Reinforced the application of academic modeling techniques to real-world infrastructure challenges under time constraints. Improved confidence in renewable energy system integration and vendor coordination.

TRANSPORTATION SYSTEM INSTALLATION SUPPORT & FIELD TROUBLESHOOTING

Client:

Regional Transportation Authority (Southern California)

Role:

Electrical & Systems Engineer – Transportation Systems Integrator

Project Overview:

Provided late-stage installation support and technical troubleshooting during final system deployment of a large-scale transportation infrastructure project. Transitioned mid-project from a production-focused role into a systems engineering position, enabling direct oversight of cabinet construction, field installation, and cross-team coordination. Addressed equipment failures, design gaps, and communication issues to maintain deployment progress and system integrity.

Key Actions and Responsibilities:

Identified structural and integration deficiencies during cabinet production and proposed corrective actions following field-discovered failures. Designed modular installation kits to complete cabinet builds and maintain continuity despite production delays. Authored standardized installation procedures and documentation for field technicians to ensure safe, consistent cabinet assembly. Managed packaging, documentation, and shipment of spare materials and equipment, including detailed inventory tracking to streamline delivery and verification at deployment sites. Collaborated with vendor engineering teams to resolve display formatting and integration issues, validating that operational messages rendered correctly on electronic signage. Supported on-site camera placement and alignment to ensure proper visual coverage and compliance with operational requirements.

Results and Impact:

Sustained project momentum despite equipment delays and design challenges. Improved coordination between engineering, production, and field teams. Delivered clear, well-structured installation packages that enhanced field efficiency and reduced rework. Increased overall system reliability and ensured final installations met design intent and operational requirements.

Key Learnings:

Reinforced the value of proactive design feedback and strong cross-department communication. Strengthened ability to bridge engineering design intent with production and field realities to improve long-term system quality and deployment efficiency.

TECHNICAL DOCUMENTATION & OPERATIONS STANDARDIZATION

Client:

Project Operations (Transportation Systems Integrator)

Role:

Electrical / Systems Engineer & Senior Production Lead

Overview:

Developed a comprehensive suite of technical documentation, Standard Operating Procedures (SOPs), and field integration guides to standardize production, installation, commissioning, and maintenance workflows across large-scale transportation system deployments. These resources were designed to reduce operational friction, improve cross-team communication, and enable consistent system implementation without reliance on ad hoc engineering support.

Technical Scope and Approach:

Authored detailed SOPs covering cabinet production workflows, workspace organization, quality control checkpoints, and progress reporting standards. Created step-by-step field technician guides to support system configuration, component replacement, and troubleshooting, with emphasis on repeatability and clarity under real-world field conditions. Developed structured firmware update procedures for roadside and cabinet-mounted devices, including interface navigation, file management, version control, and validation steps. Consolidated all required software, firmware, drivers, and configuration tools into a single deployable package to streamline field use and eliminate version mismatch issues.

Produced standardized design references including wiring diagrams, configuration sheets, annotated screenshots, and component cut sheets to ensure consistent interpretation across engineering, production, and field teams.

Key Deliverables:

- Standard Operating Procedures for production, installation, and maintenance workflows
- Field-ready technical guides for system setup, configuration, and troubleshooting
- Structured firmware update and device configuration procedures
- Centralized deployment package containing required software, firmware, and drivers
- Reference documentation including wiring diagrams, design sheets, and hardware summaries

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TECHNICAL DOCUMENTATION & OPERATIONS STANDARDIZATION

Results and Impact:

Standardized documentation across teams improved consistency, reduced installation errors, and accelerated troubleshooting timelines. Training efficiency increased significantly, shortening onboarding time for new technical staff and reducing dependence on direct engineering oversight. Field teams were able to independently complete system setup, verification, and maintenance tasks with minimal escalation, improving overall operational resilience and scalability.

Key Learnings:

Reinforced the importance of proactive documentation as both an operational backbone and a long-term training asset. Strengthened technical communication skills and demonstrated how structured knowledge transfer enables team autonomy, reduces risk during personnel transitions, and supports scalable system deployment and maintenance.

INDEPENDENT SYSTEMS & CONTROLS DEVELOPMENT (SELF-STUDY)

In addition to production and field-deployed work, I actively pursue independent systems and controls development to deepen my understanding of plant-level control architecture, SCADA design, and grid-facing behavior.

My public GitHub repository documents a self-directed Solar Plant SCADA and Power Plant Controller study implemented in CODESYS. The project models a utility-scale solar-plus-storage facility operating under an ISO-style dispatch paradigm, including ramp enforcement, asset coordination, energy constraints, and operator-facing SCADA visibility. The repository includes the full control logic, simulation models, SCADA configuration, system design documentation, and a structured verification and acceptance test presentation.

This work reflects how I approach unfamiliar or adjacent domains: by building complete, testable systems that emphasize control authority, deterministic behavior, and operational clarity. It is intended as a technical demonstration of control philosophy and system architecture rather than a claim of certification or market participation.

GitHub Repository:

<https://github.com/hornjacob99-beep/Solar-Plant-SCADA-System>